Research Plan: Dynamic Multi-Sensor GeoAI Framework for Real-Time Landslide Susceptibility Evolution in Nepal

IOCGM 2027 Researcher Skill Output

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1 Introduction

This research plan outlines a comprehensive workflow to model landslide susceptibility in the Arun River Basin, Nepal, using 2020–2026 data. The core innovation lies in the **Hybrid Rainfall Fusion** (DHM + GPM) and the **Sentinel-1/2 Synergy**.

2 Research Gaps & Literature Review

- **Temporal Gap:** Existing maps (Durham/BGS) end in 2020. This study covers the 2021–2026 era.
- **Data Gap:** Most models use only optical data. We utilize Sentinel-1 SAR as a moisture proxy.
- **Explainability:** Shifting from "black-box" ML to SHAP-based Explainable AI (XAI).

3 Methodology

3.1 Hybrid Rainfall Fusion

We address DHM Nepal data gaps (due to COVID-19/Earthquake legacy) by performing a linear bias correction on GPM IMERG satellite data using available DHM station records.

$$P_{fused} = \begin{cases} P_{DHM} & \text{if data exists} \\ \alpha \cdot P_{GPM} & \text{to fill gaps} \end{cases} \tag{1}$$

3.2 Modeling

We employ an **XGBoost** ensemble with an ablation study:

- Model A: Static Topography
- Model B: Topography + S2 (NDVI)
- Model C: Topography + S2 + S1 (Full Synergy)

4 Timeline

Month 1: Data Curation, Hybrid Rainfall Fusion, GEE Feature Extraction.

Month 2: Model Training, Ablation Study, SHAP Analysis.

Month 3: Cross-Basin Transferability (Trishuli), Abstract Finalization.

5 Computation Time Estimates

- GEE Feature Extraction: 2–4 hours (Cloud-based).
- Local Processing (CSV): 1–2 hours (on a standard laptop).